Experiment 6 : Data Aggregation

Aim: To analyze and visualize retail sales data by aggregating and comparing sales across regic product categories.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
np.random.seed(5)
n = 400
df = pd.DataFrame({
    "date": pd.to_datetime(np.random.choice(pd.date_range("2026-01-01", p
    "region": np.random.choice(["North", "South", "East", "West"], n),
    "category": np.random.choice(["Electronics", "Clothing", "Grocery", "
    "quantity": np.random.randint(1, 10, n),
    "amount": np.round(np.random.uniform(100, 5000, n), 2),
})
df.head()
```

Out[2]:

	date	region	category	quantity	amount
0	2026-04-10	South	Grocery	6	597.37
1	2026-04-29	West	Electronics	5	2115.81
2	2026-05-25	South	Grocery	3	4366.17
3	2026-03-15	East	Grocery	1	1099.17
4	2026-01-09	South	Electronics	7	1825.91

In [3]:
```
print("total rows:", df.shape)
```

```
total rows: (400, 5)
```

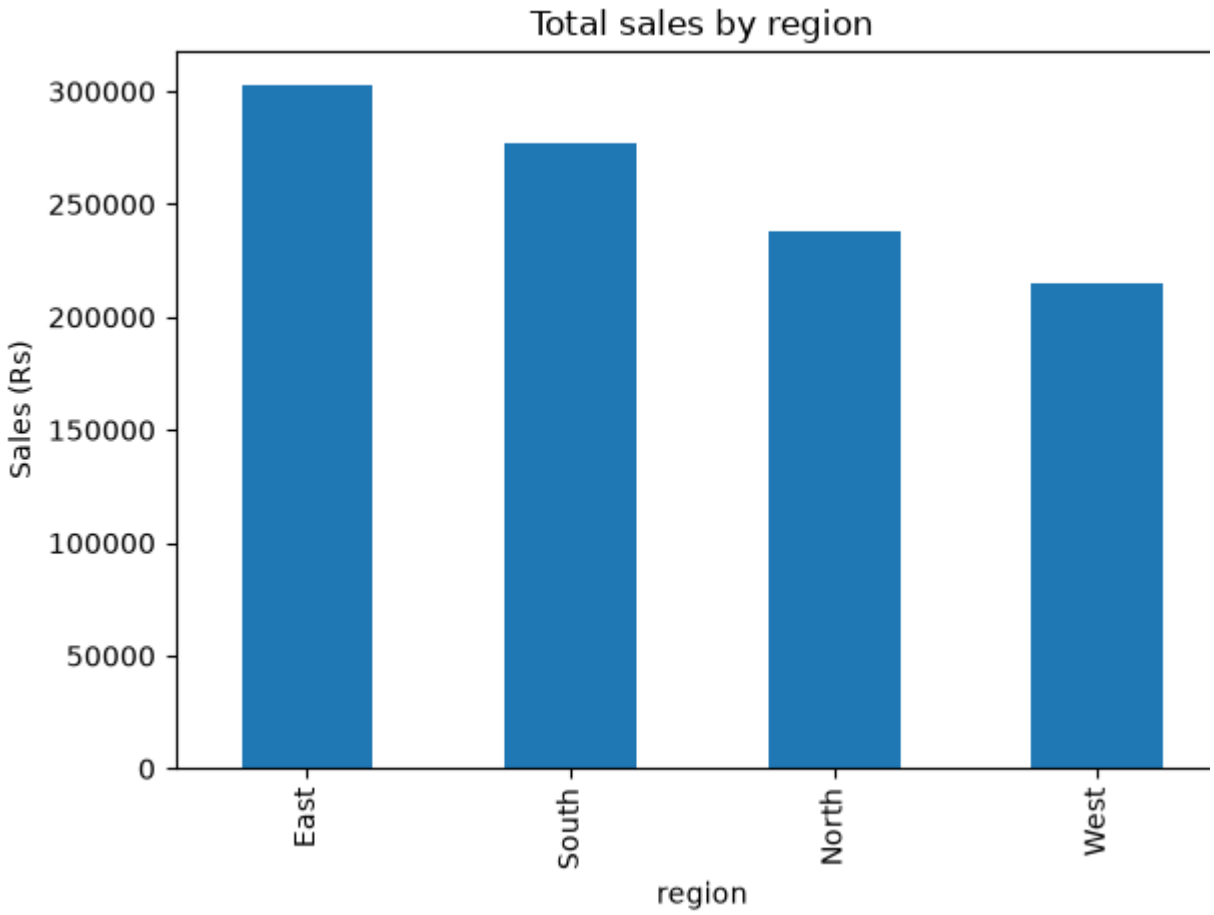
In [4]:
```
by_region = df.groupby("region")["amount"].sum().sort_values(ascending=Fa
by_region
```

Out[4]:
```
region
East     302415.93
South    277106.73
North    238128.91
West     214866.47
Name: amount, dtype: float64
```

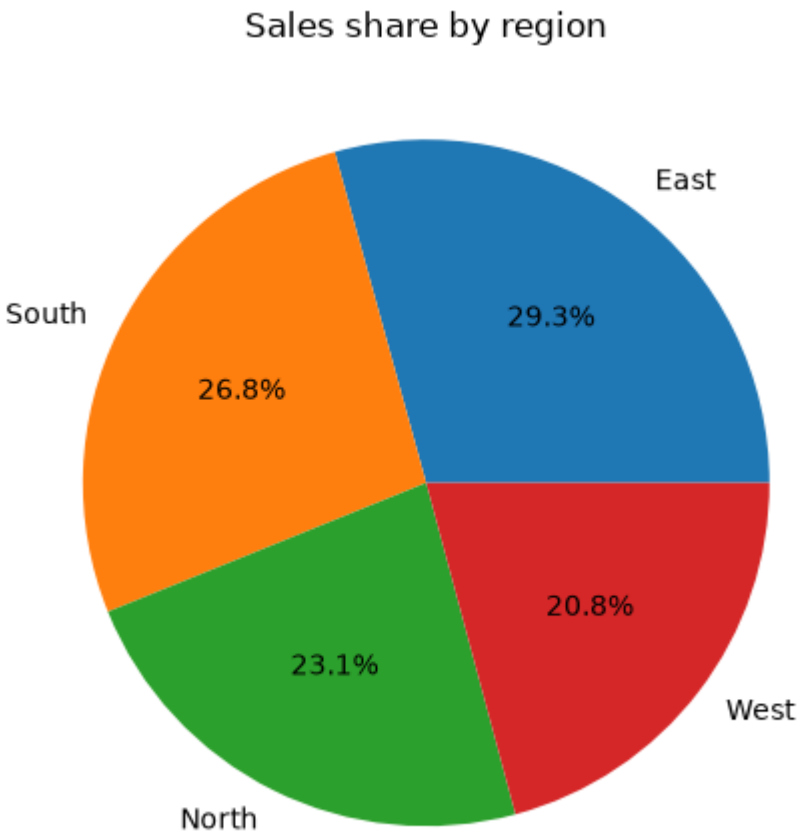
In [5]:
```
print("Top region:", by_region.index[0])
```

```
Top region: East
```

In [6]:
```
by_region.plot(kind="bar")
plt.title("Total sales by region")
plt.ylabel("Sales (Rs)")
plt.tight_layout()
plt.show()
```



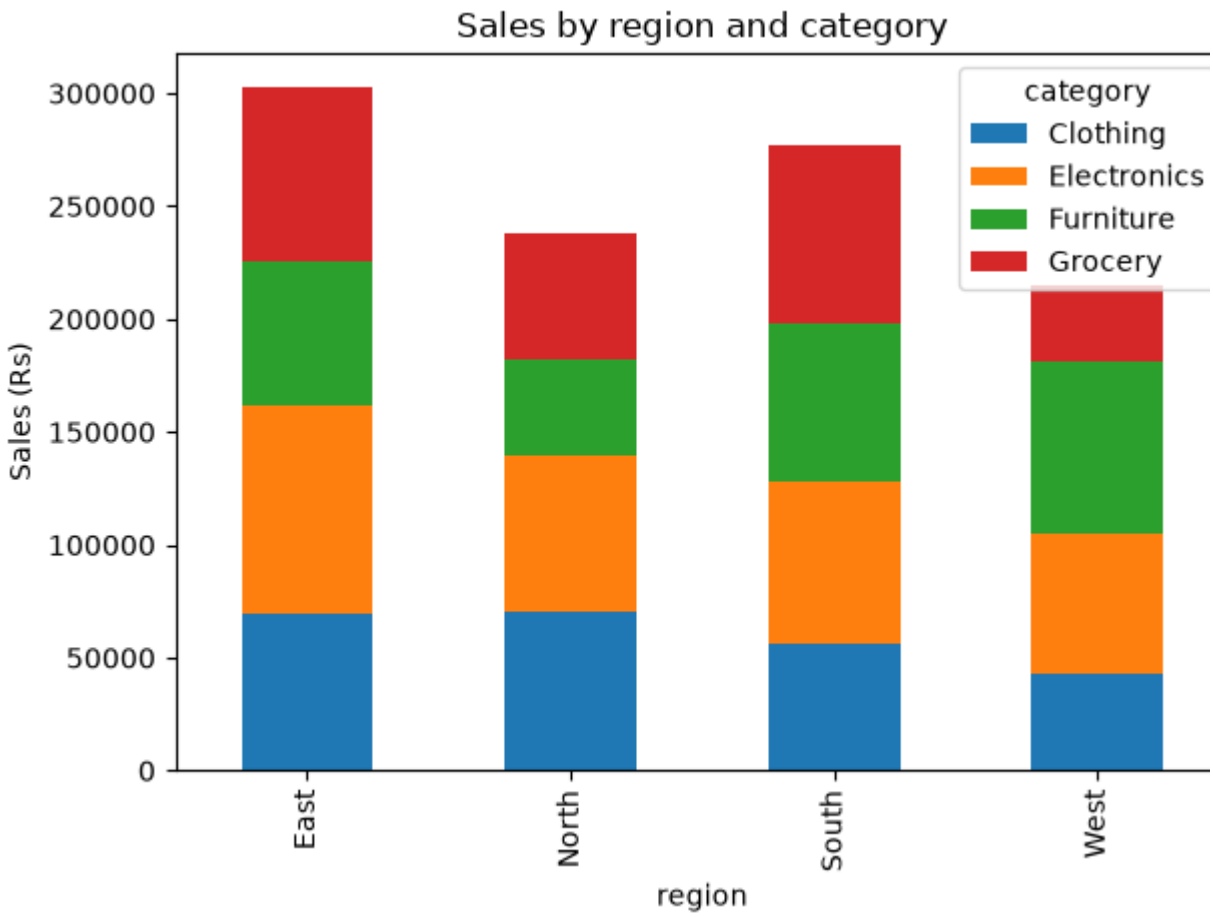
In [7]:
```
by_region.plot(kind="pie", autopct="%1.1f%%")
plt.title("Sales share by region")
plt.ylabel("")
plt.tight_layout()
plt.show()
```



In [8]:
```
by_both = df.groupby(["region", "category"])["amount"].sum().unstack()
print(by_both)
```

```
category  Clothing  Electronics  Furniture   Grocery
region
East      69210.35     92355.88   64309.44  76540.26
North     70766.43     68569.85   42383.33  56409.30
South     56336.63     71520.16   70409.72  78840.22
West      42871.62     62501.51   75803.58  33689.76
```

In [9]:
```
by_both.plot(kind="bar", stacked=True)
plt.title("Sales by region and category")
plt.ylabel("Sales (Rs)")
plt.tight_layout()
plt.show()
```

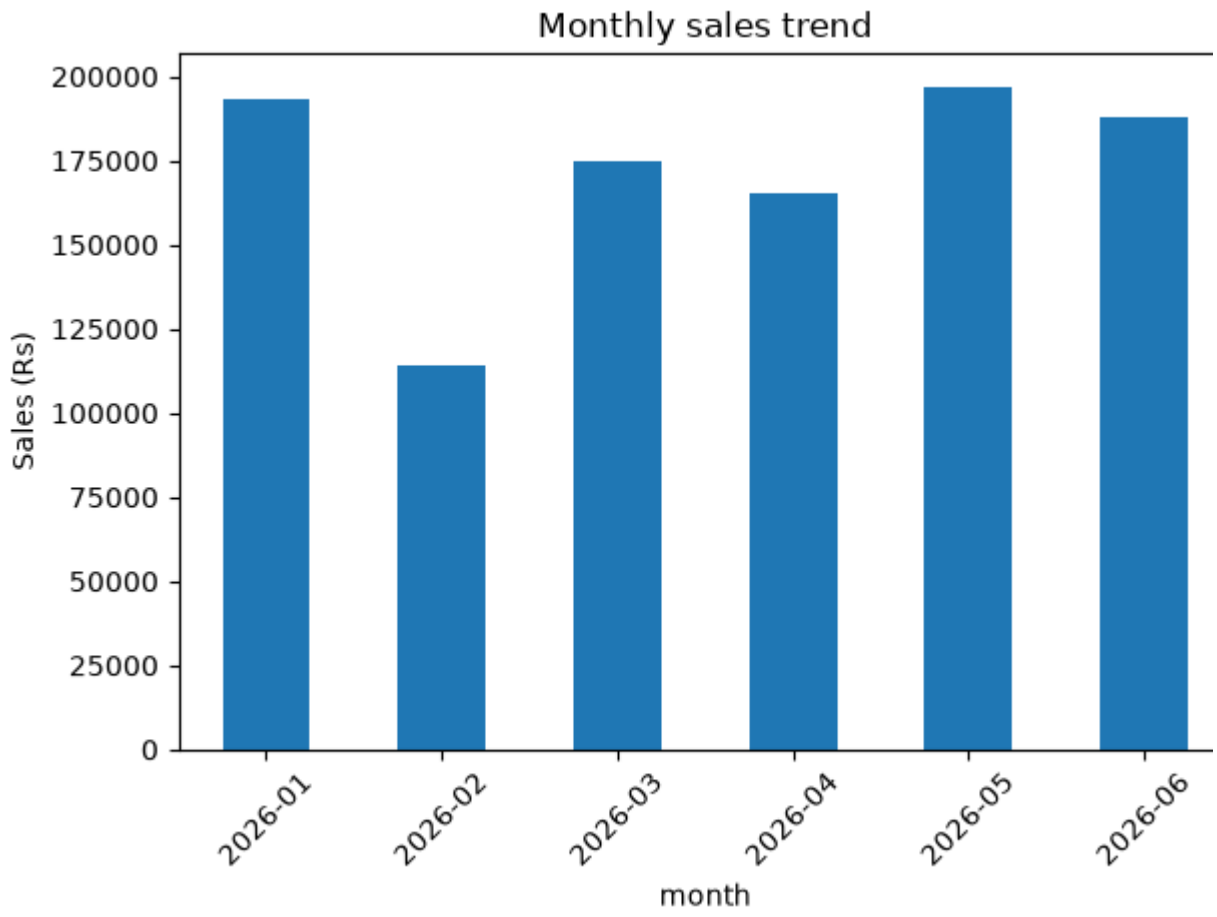


In [10]:
```
df["month"] = df["date"].dt.to_period("M").astype(str)
df.head()
```

Out[10]:

	date	region	category	quantity	amount	month
0	2026-04-10	South	Grocery	6	597.37	2026-04
1	2026-04-29	West	Electronics	5	2115.81	2026-04
2	2026-05-25	South	Grocery	3	4366.17	2026-05
3	2026-03-15	East	Grocery	1	1099.17	2026-03
4	2026-01-09	South	Electronics	7	1825.91	2026-01

In [11]:
```
df.groupby("month")["amount"].sum().plot(kind="bar")
plt.title("Monthly sales trend")
plt.ylabel("Sales (Rs)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [12]:
```
print("done")
```

```
done
```